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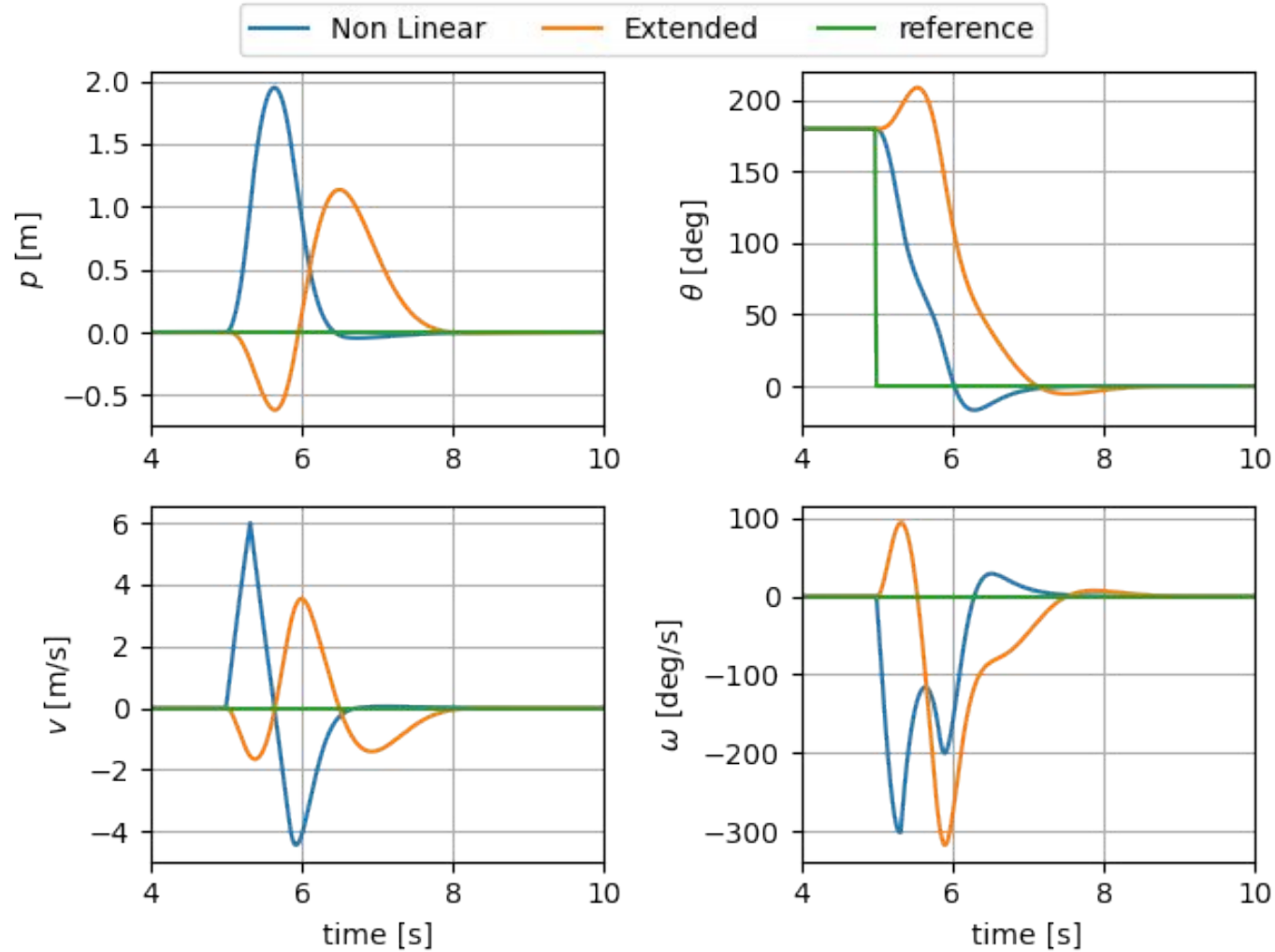
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Model Predictive Control

Comparison of Different Control Techniques for the
Inverted Pendulum Model

Lorenzo Trubian
ID 2125512

Extended Controller

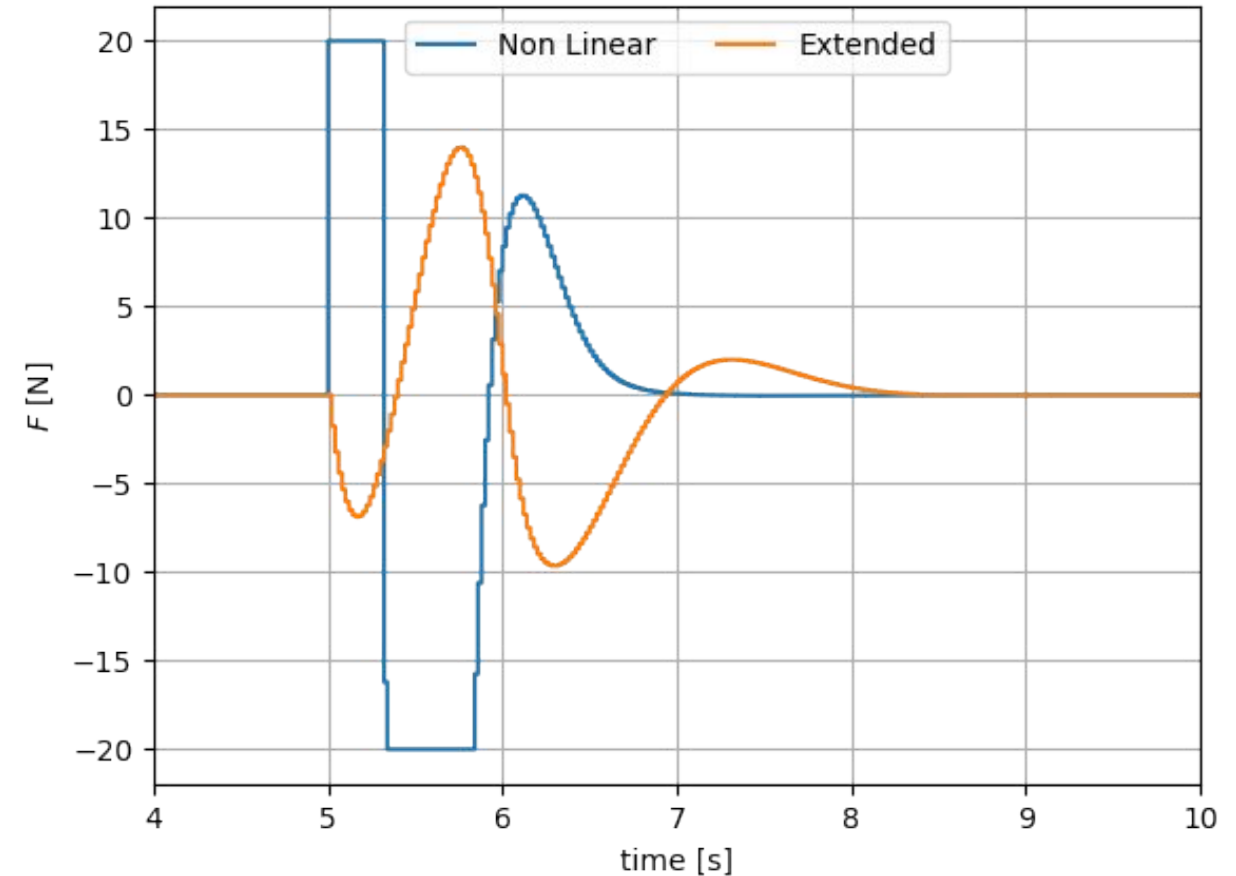
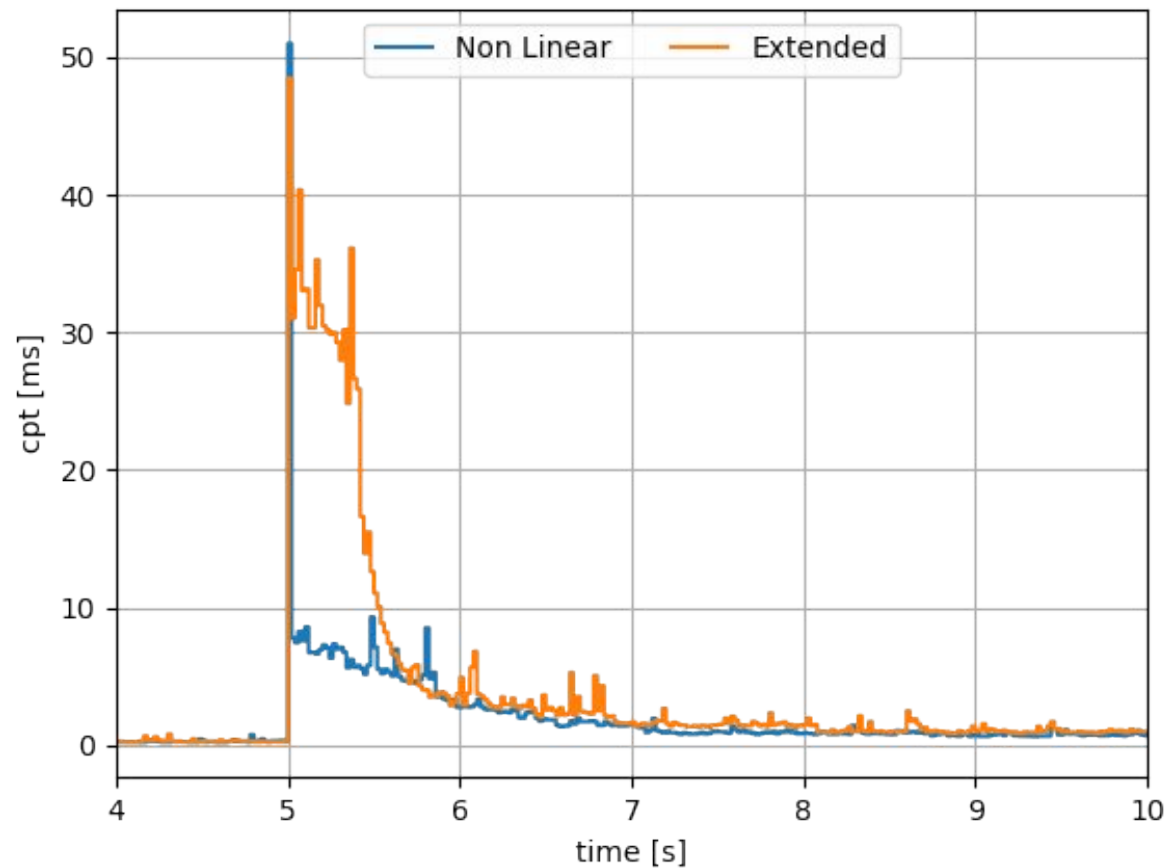


Effects on the states performances

	Non Linear	Extended
Overshoot [%]	9.31	3.01
Undershoot [deg]	0	28.8
Settling Time [s]	1.92	2.89
Mean CPU time	1	~ 1.87

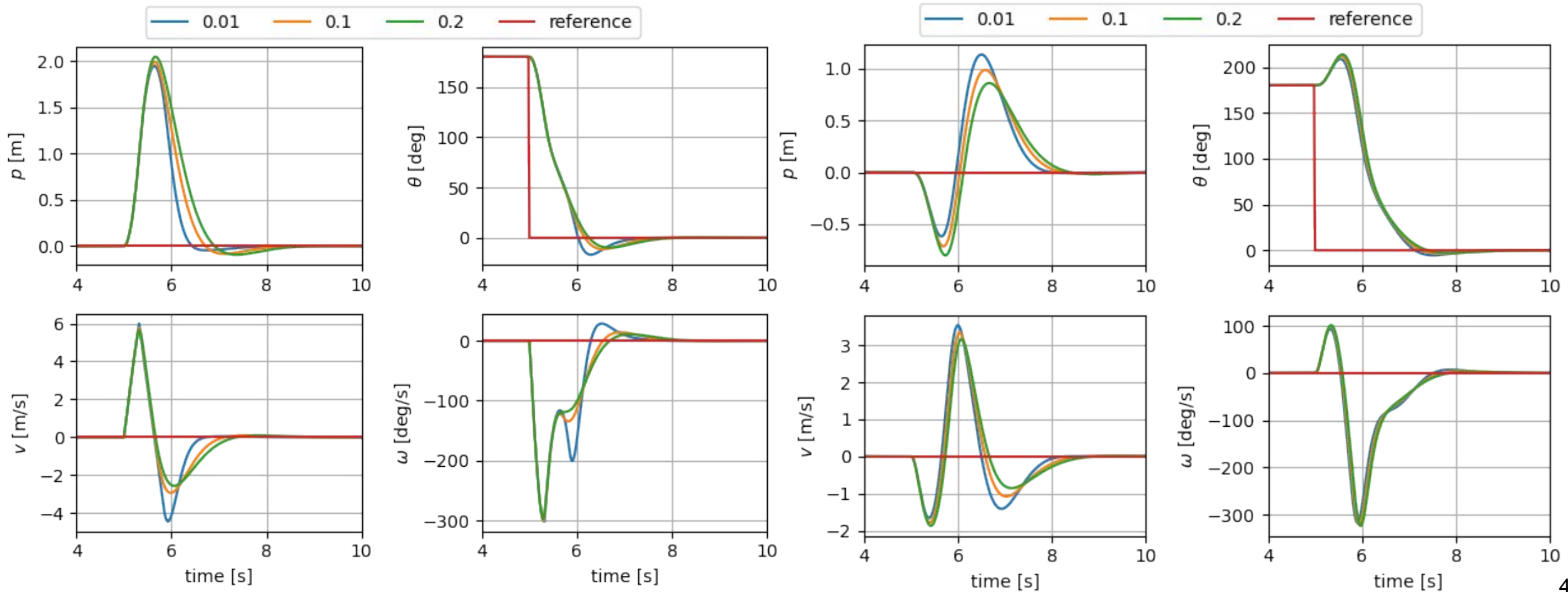
Extended Controller

Effects on the amplitude of the input and the CPU time



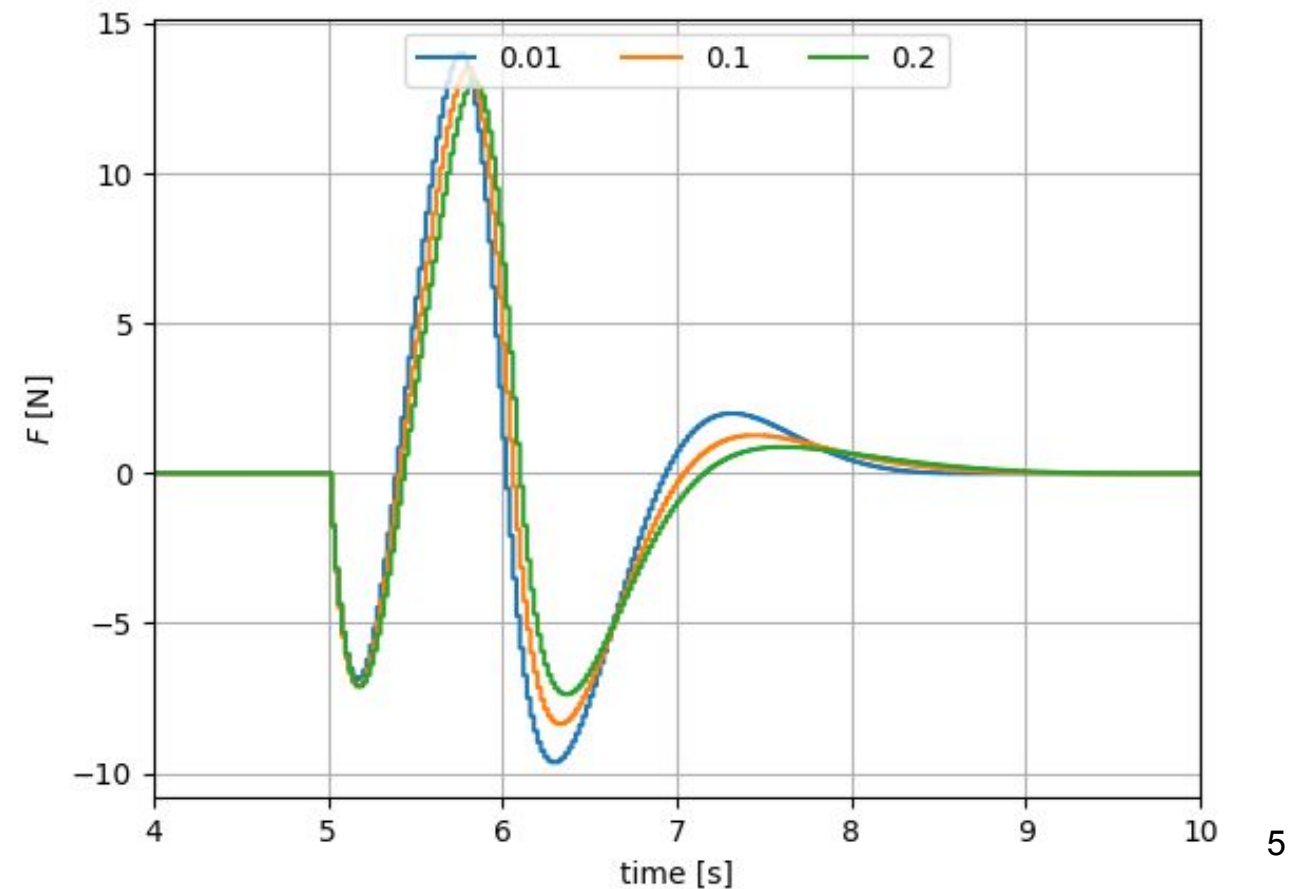
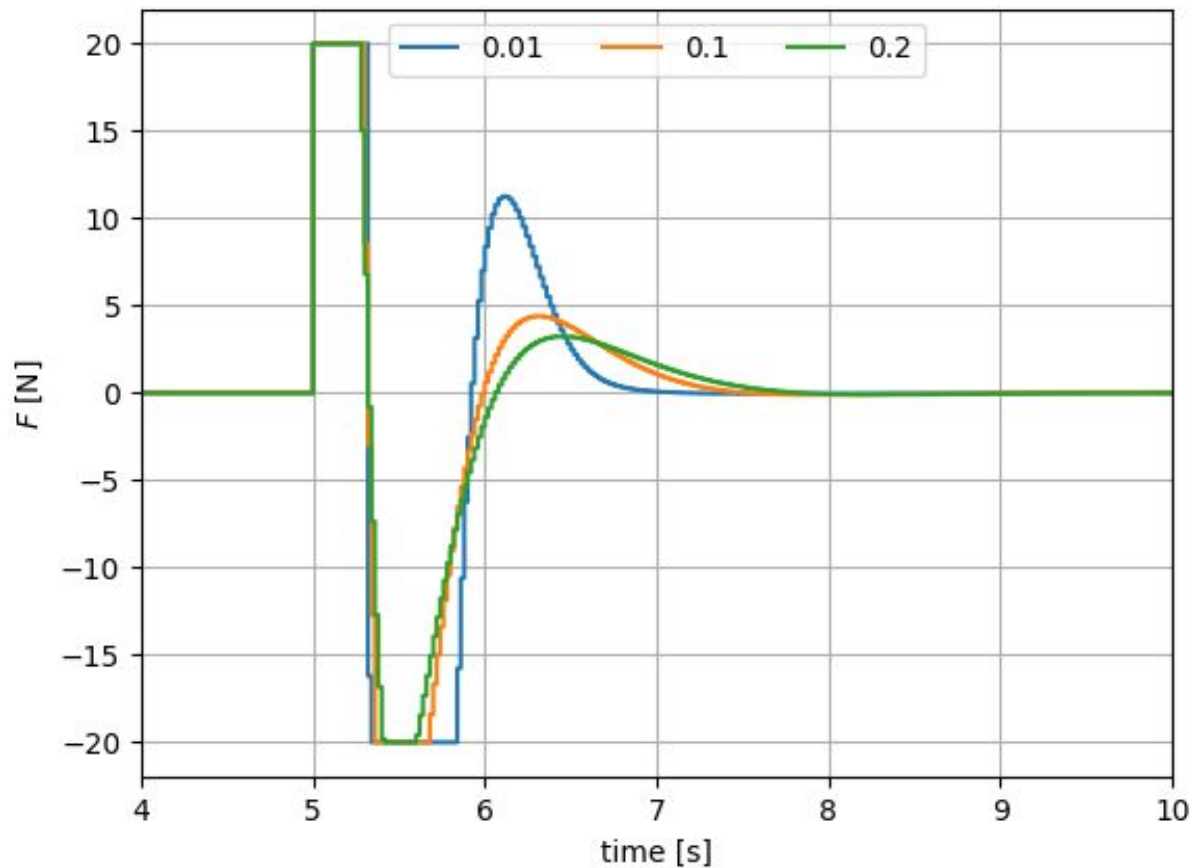
Extended Controller

Comparison of the states varying the weight corresponding to the input of the system



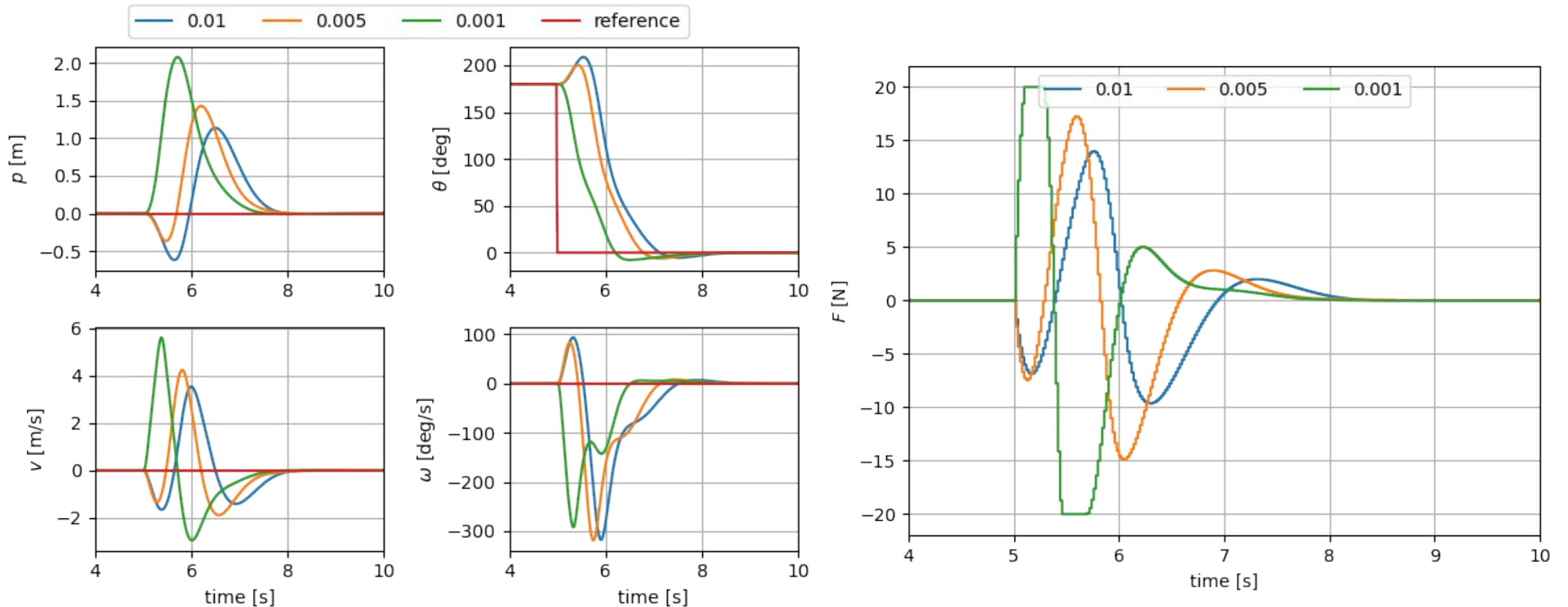
Extended Controller

Comparison of the inputs varying the weight corresponding to the input of the system



Extended Controller

Comparison of the states and inputs varying the weight corresponding to the rate of the input of the system

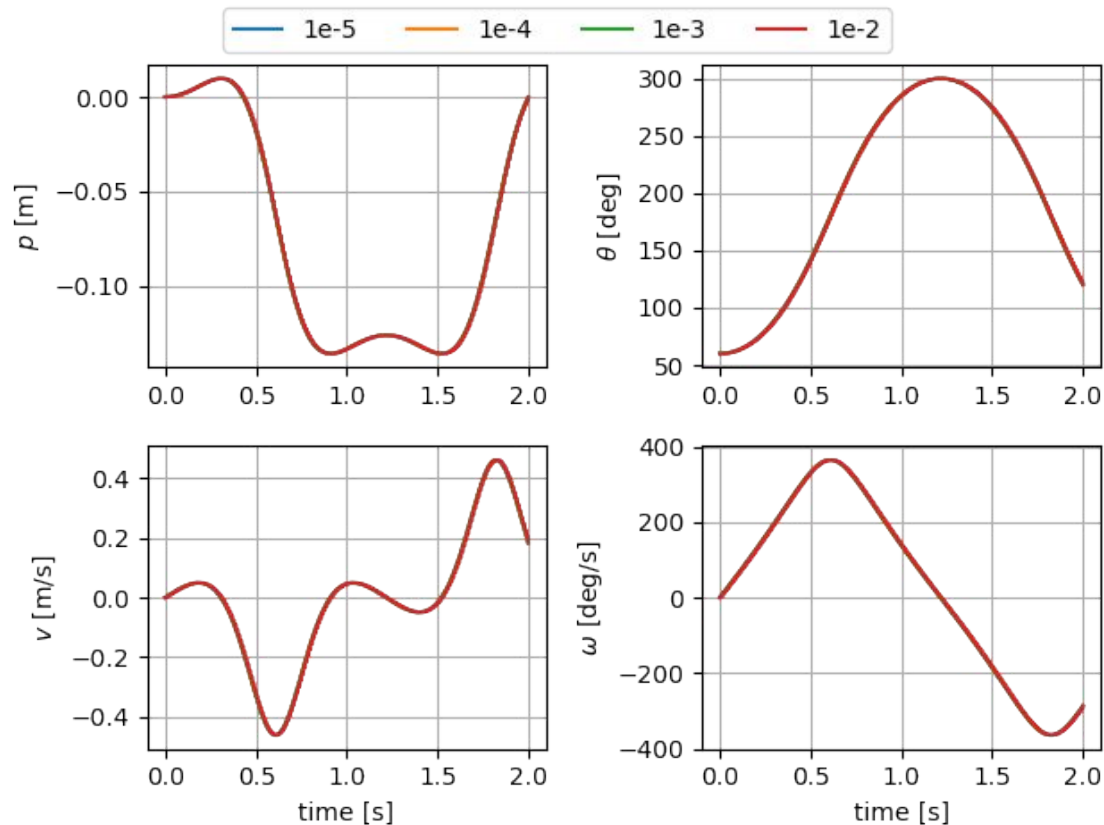


Solver Integrator Analysis on the original system

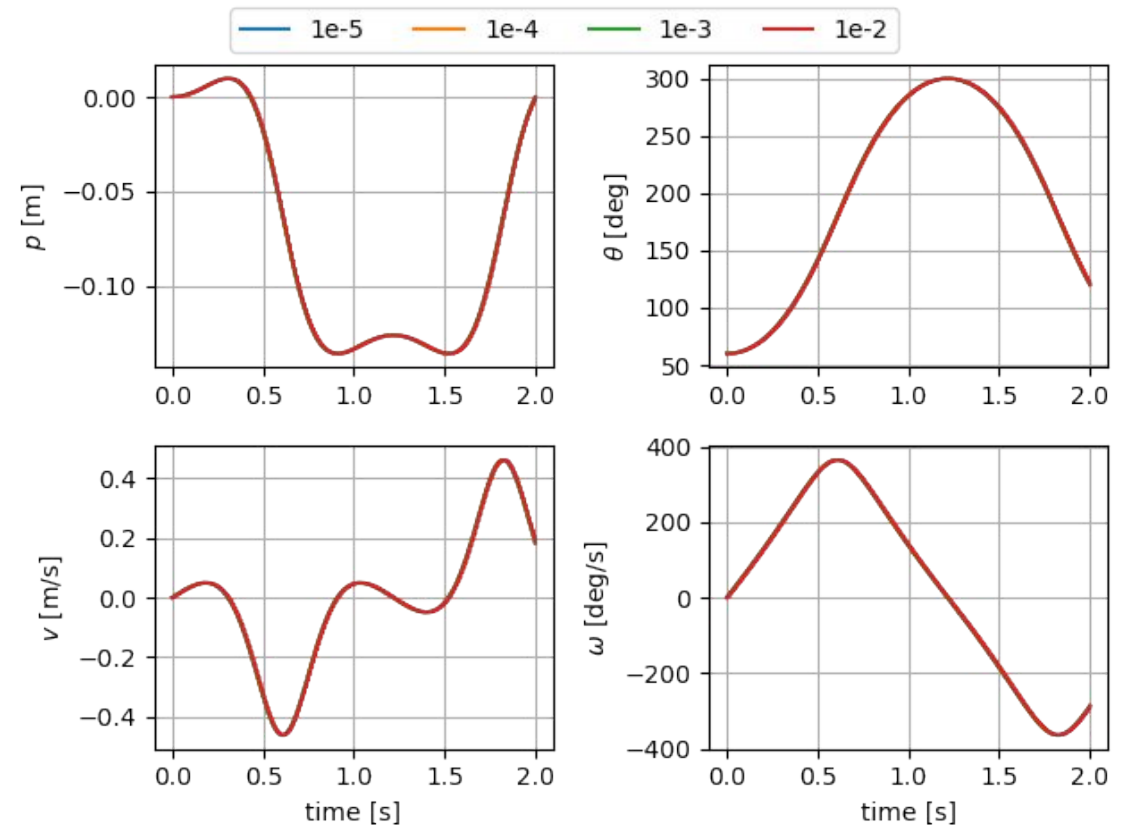
Free evolution of the original system varying the solver integrator

	ERK	IRK
Mean CPU time	1	~ 8.37

ERK



IRK

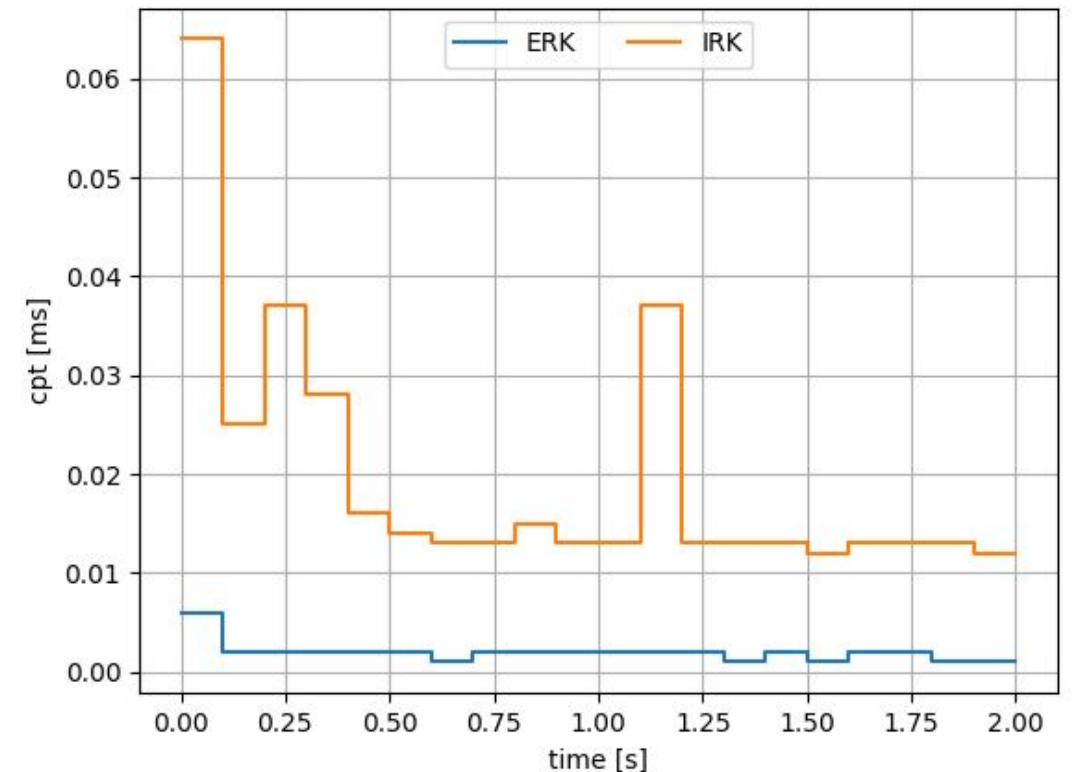
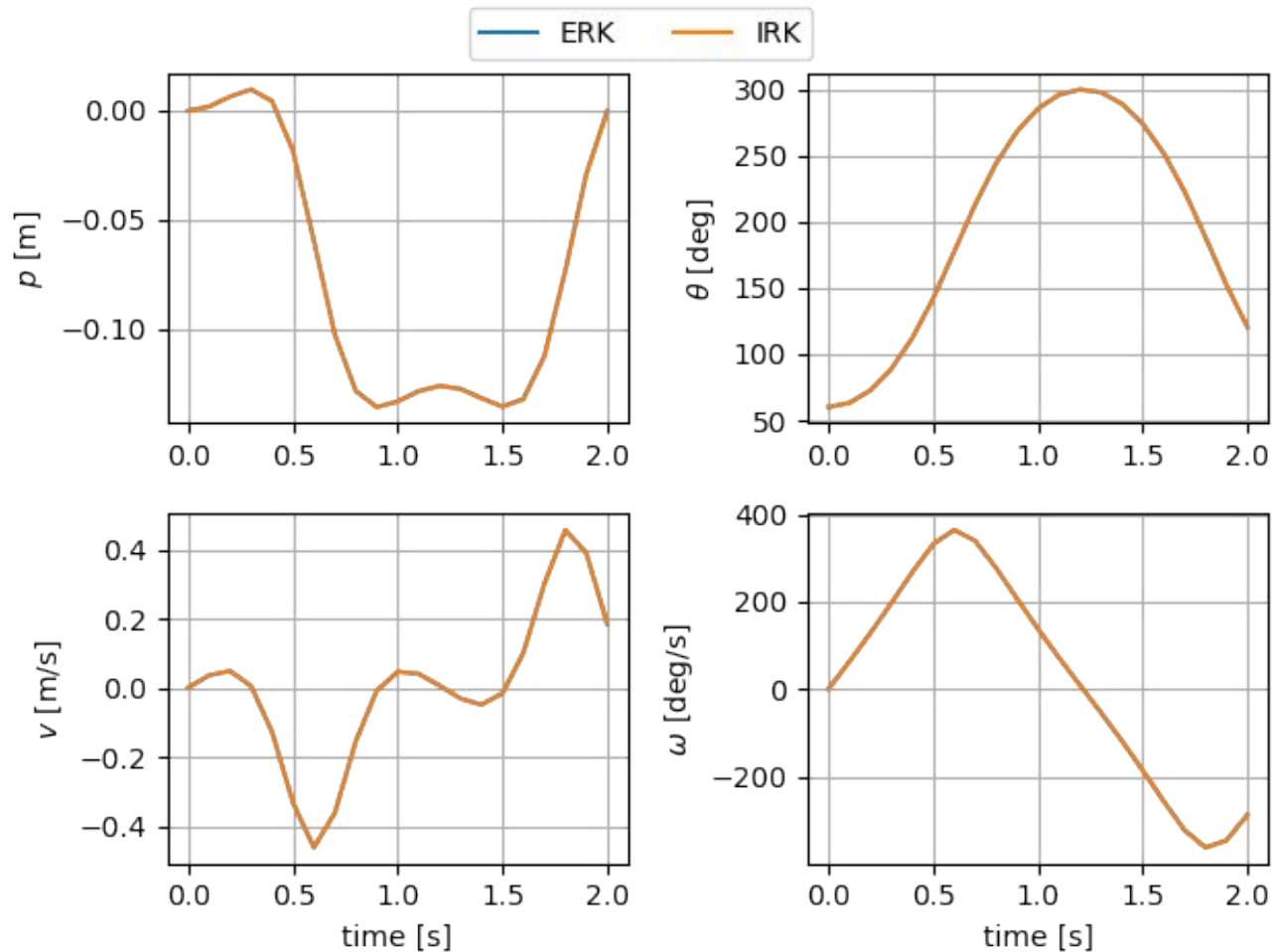


Solver Integrator Analysis on the original system

Free evolution of the original system varying the solver integrator

$T_s \text{ sim} = 0.1$

	ERK	IRK
Mean CPU time	1	~ 9.35

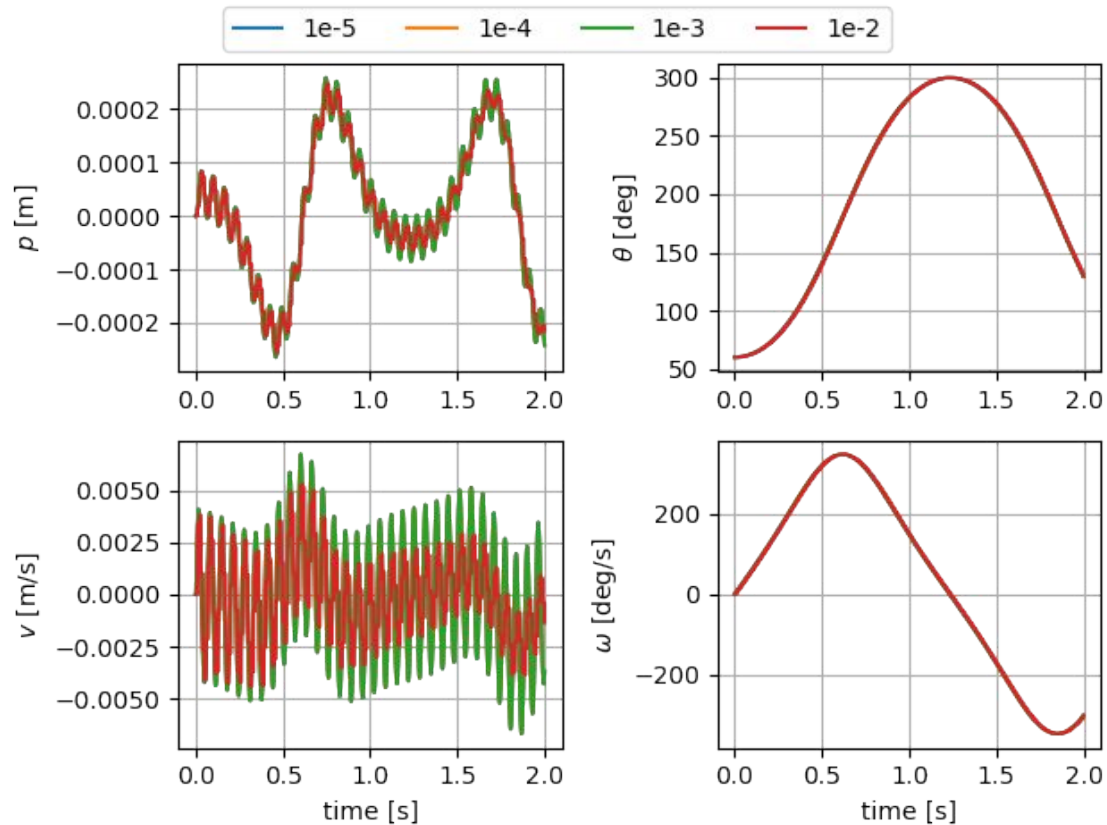


Solver Integrator Analysis on the stiff system

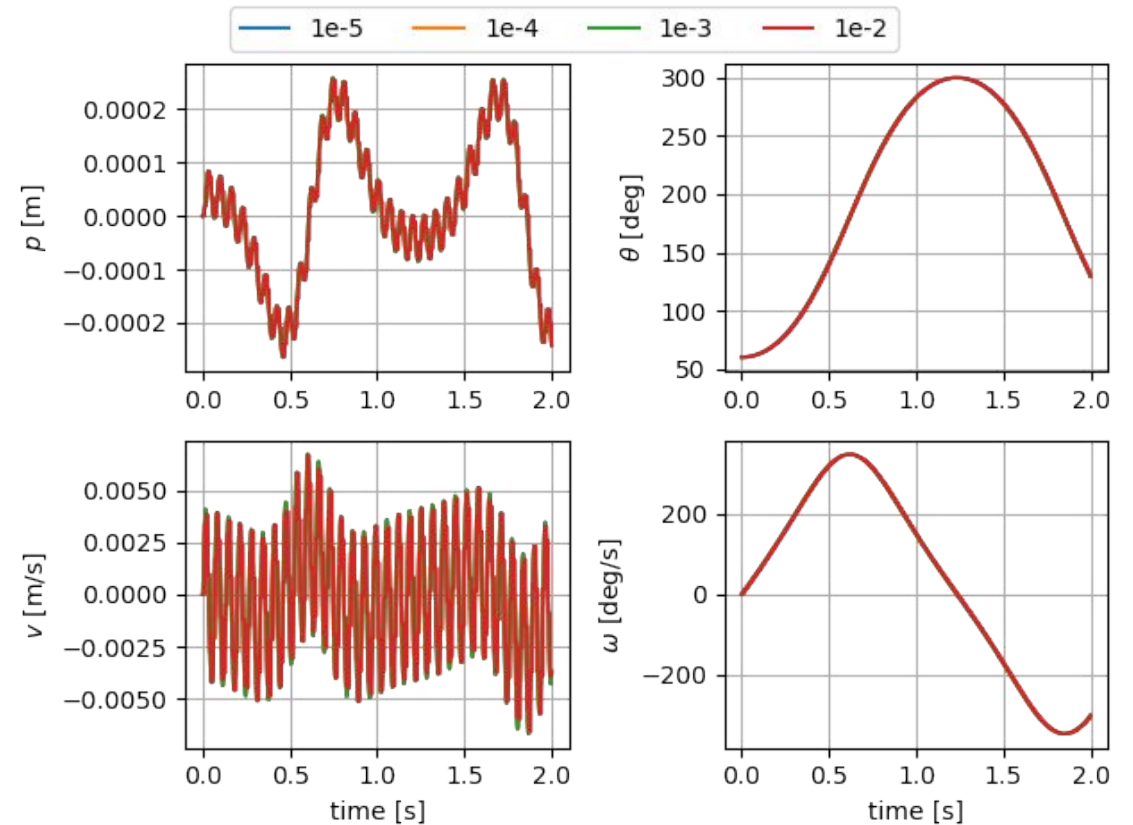
Free evolution of the stiff system varying the solver integrator

	ERK	IRK
Mean CPU time	1	~ 9.24

ERK



IRK

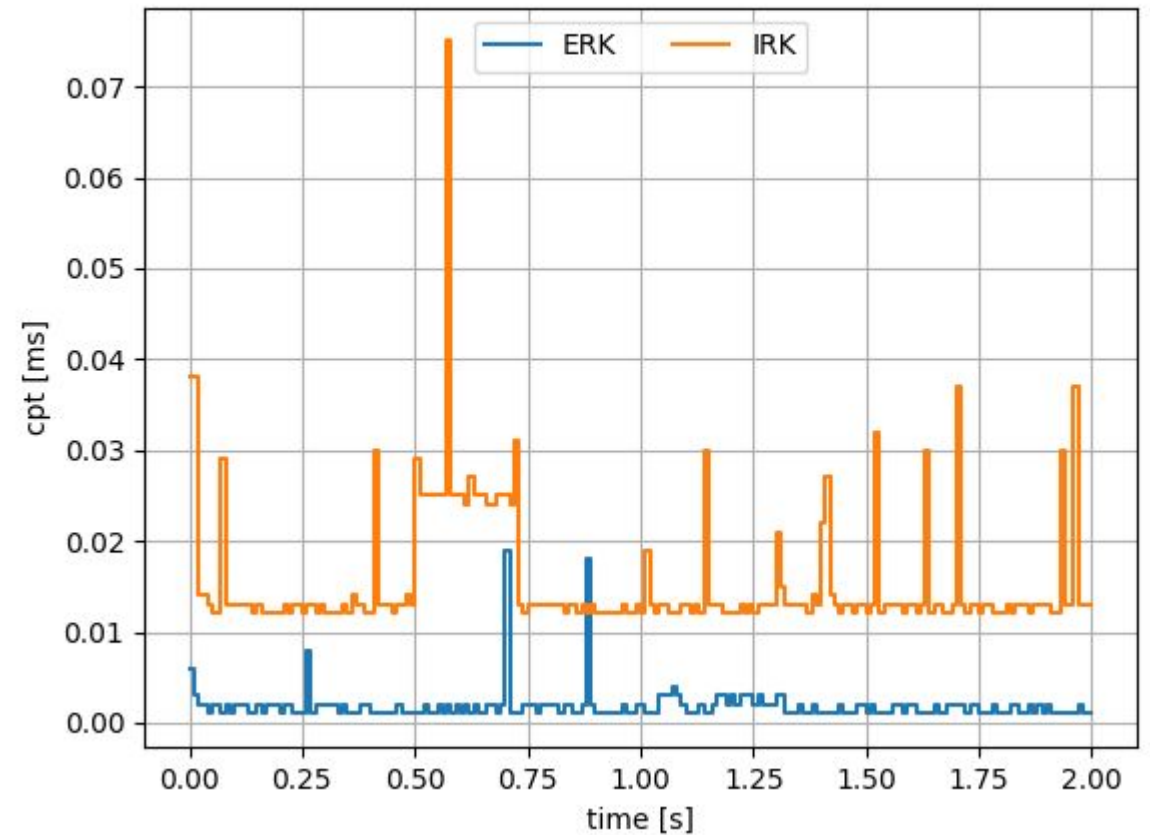
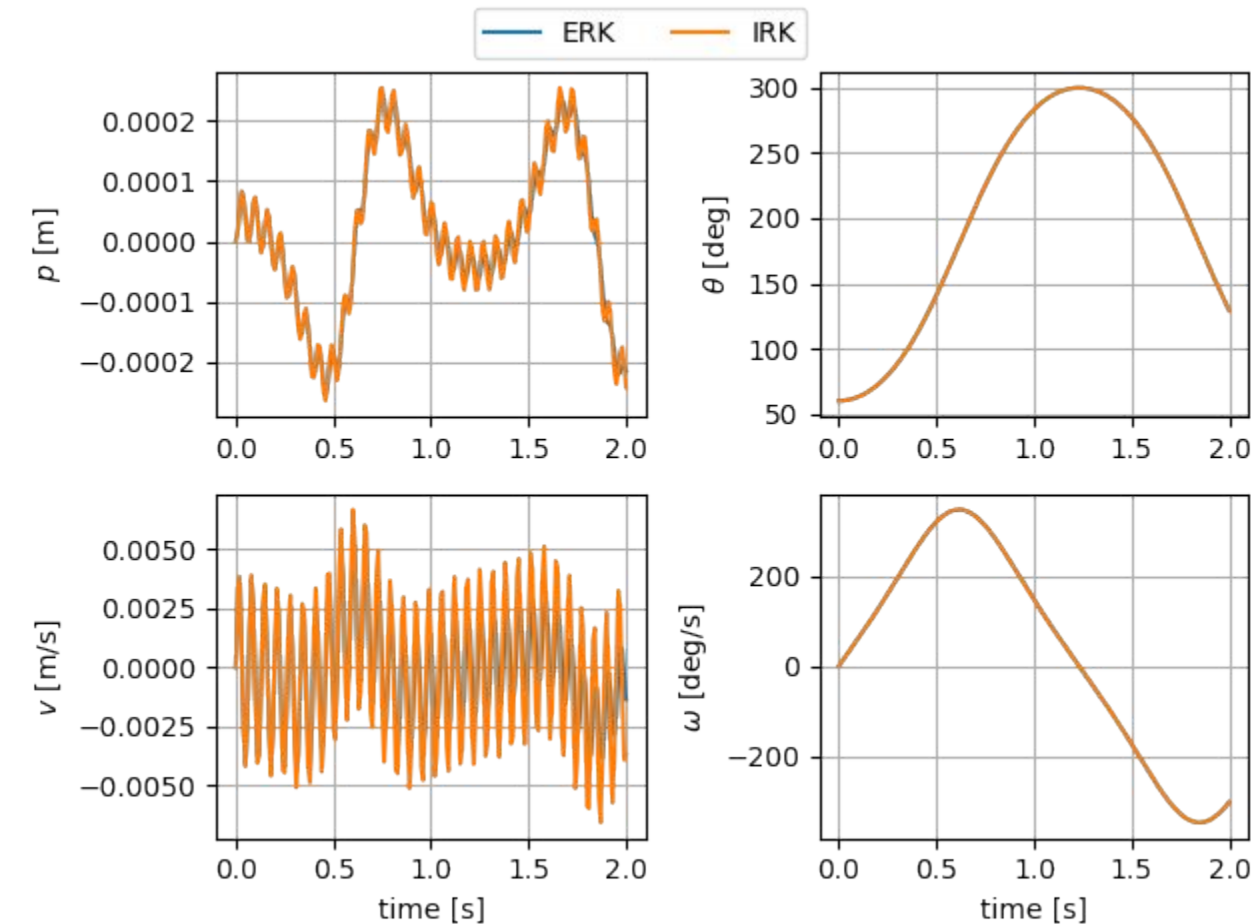


Solver Integrator Analysis on the stiff system

Comparison of the states and CPU time varying simulation integrator

$T_s \text{ sim} = 0.01$

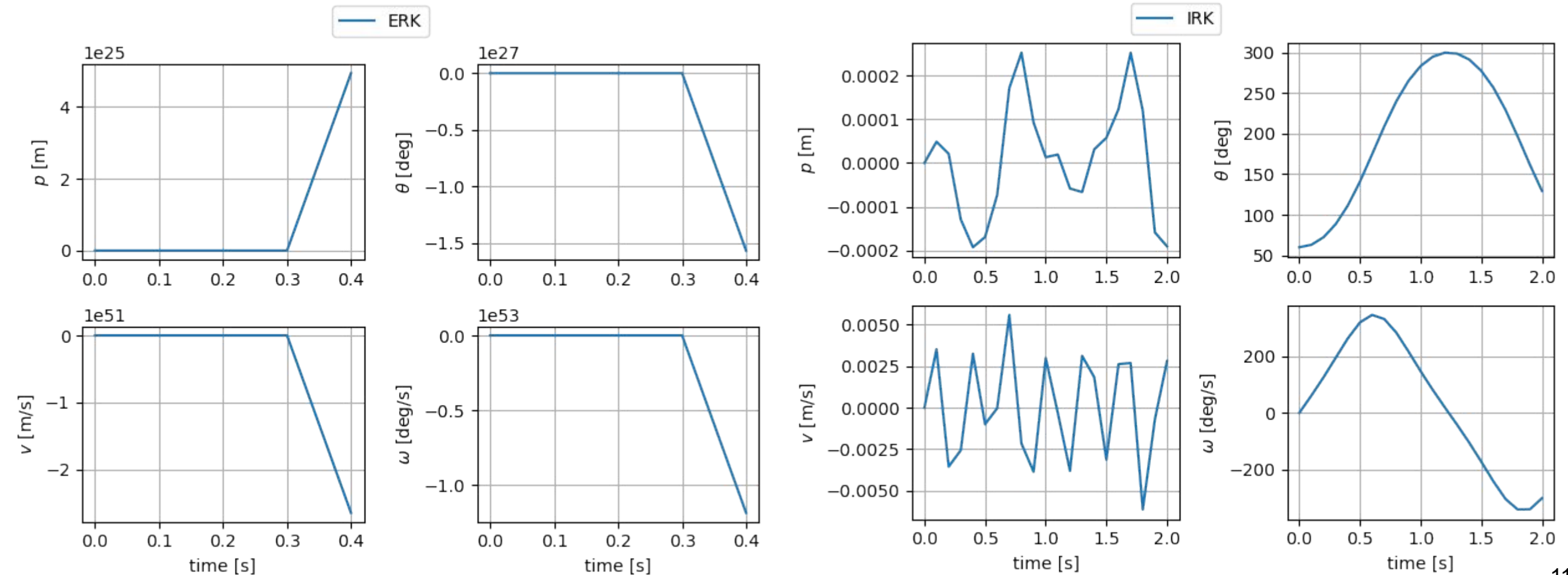
	ERK	IRK
Mean CPU time	1	~ 8.43



Solver Integrator Analysis on the stiff system

Comparison of the states varying simulation integrator

$T_s \text{ sim} = 0.01$

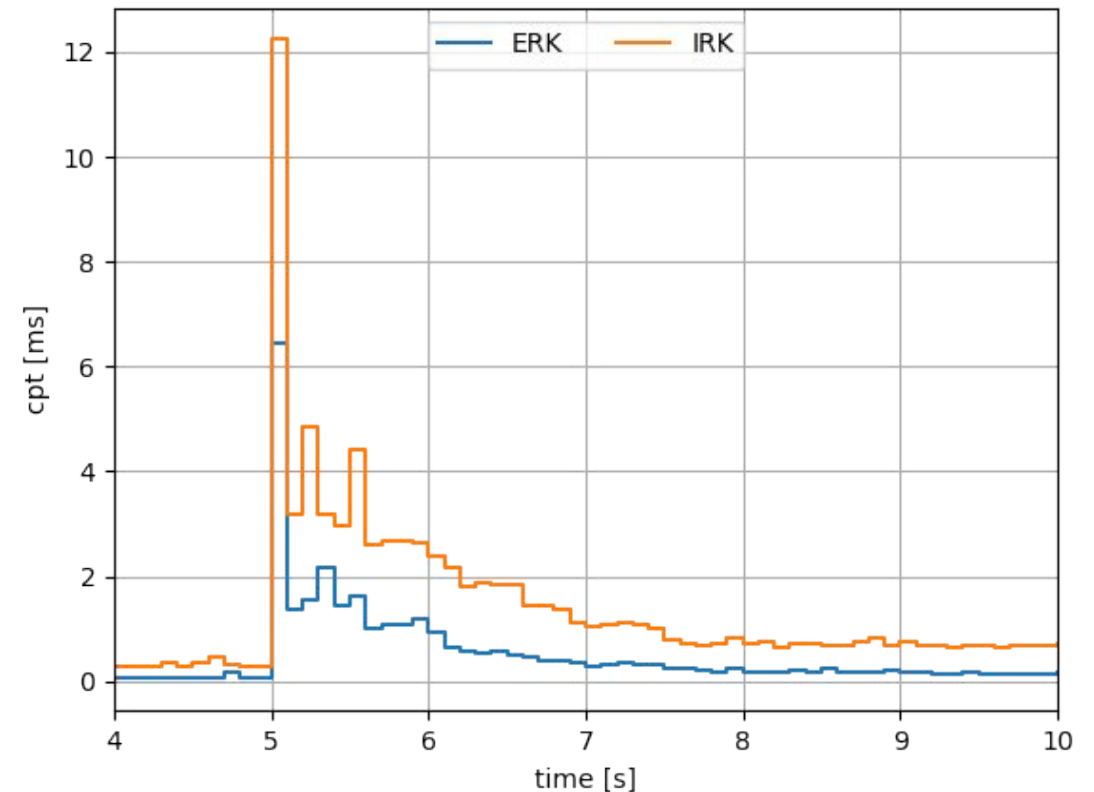
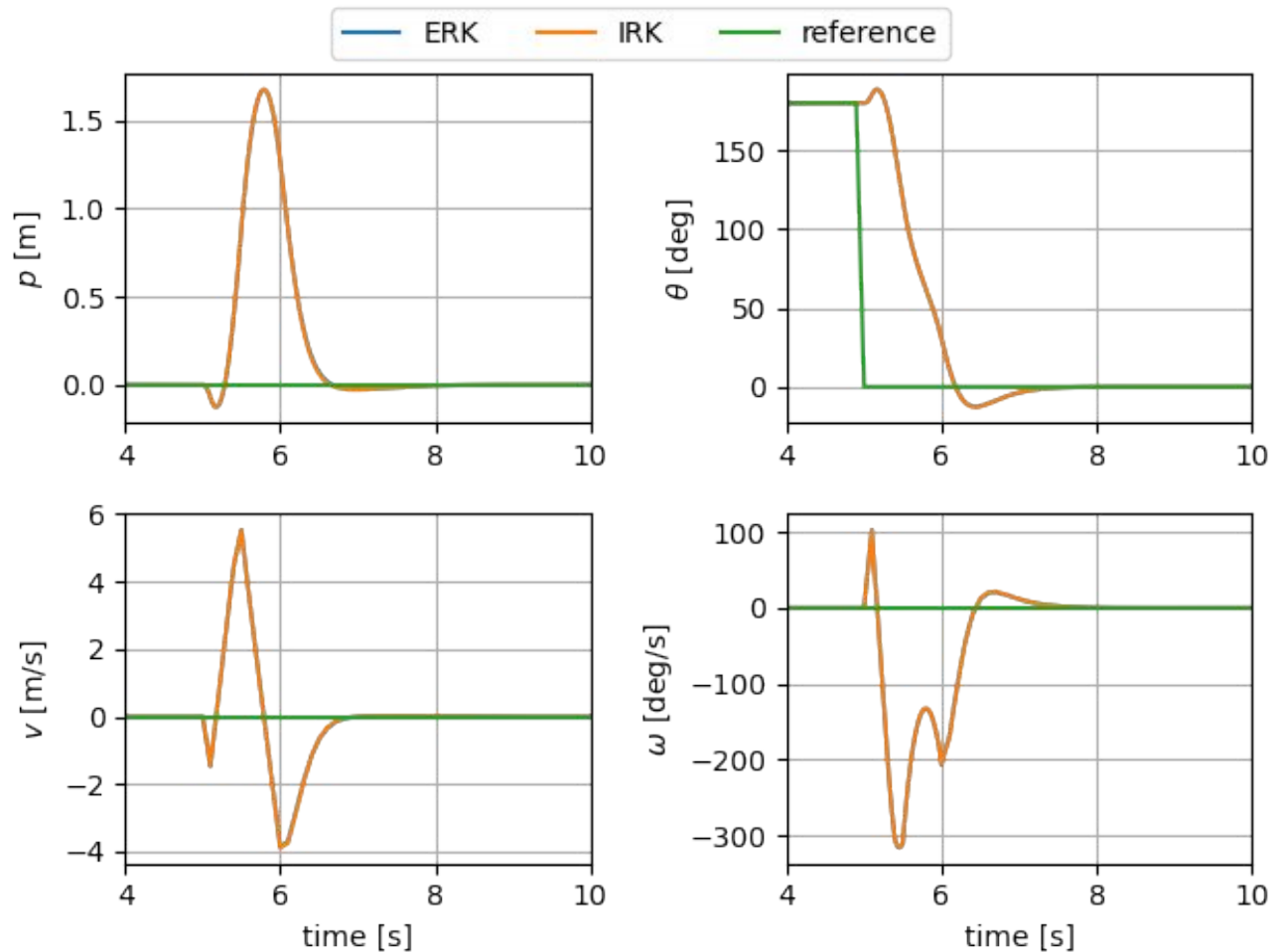


Controller Integrator Analysis

Comparison of the states and CPU time varying solver integrator

	ERK	IRK
Mean CPU time	1	~ 3.18

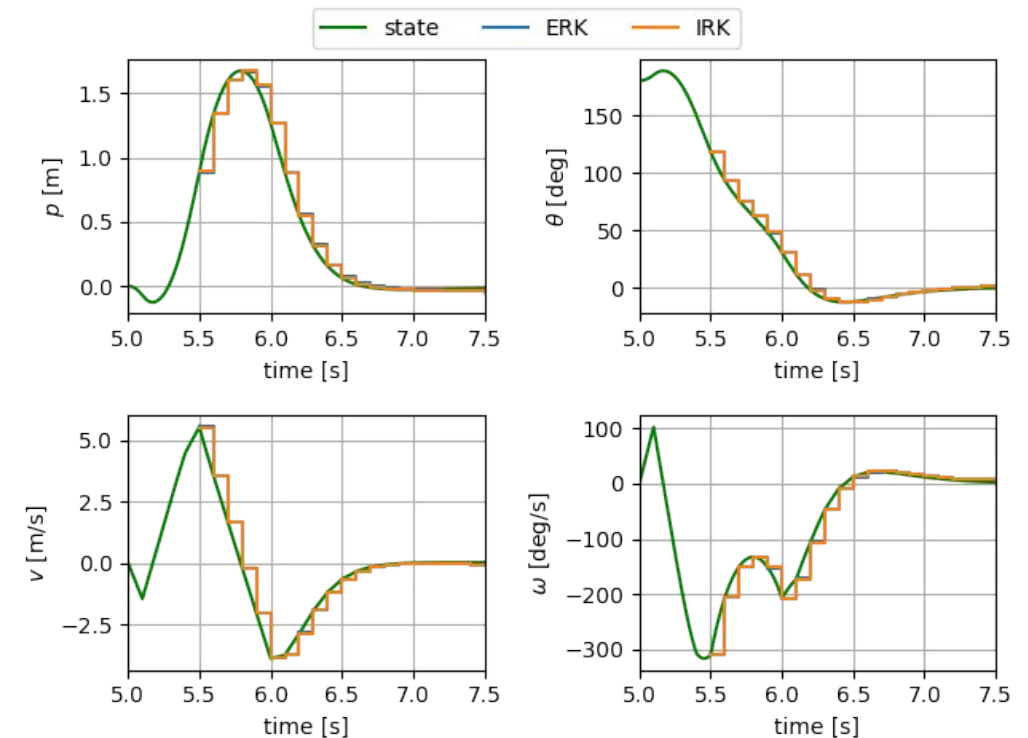
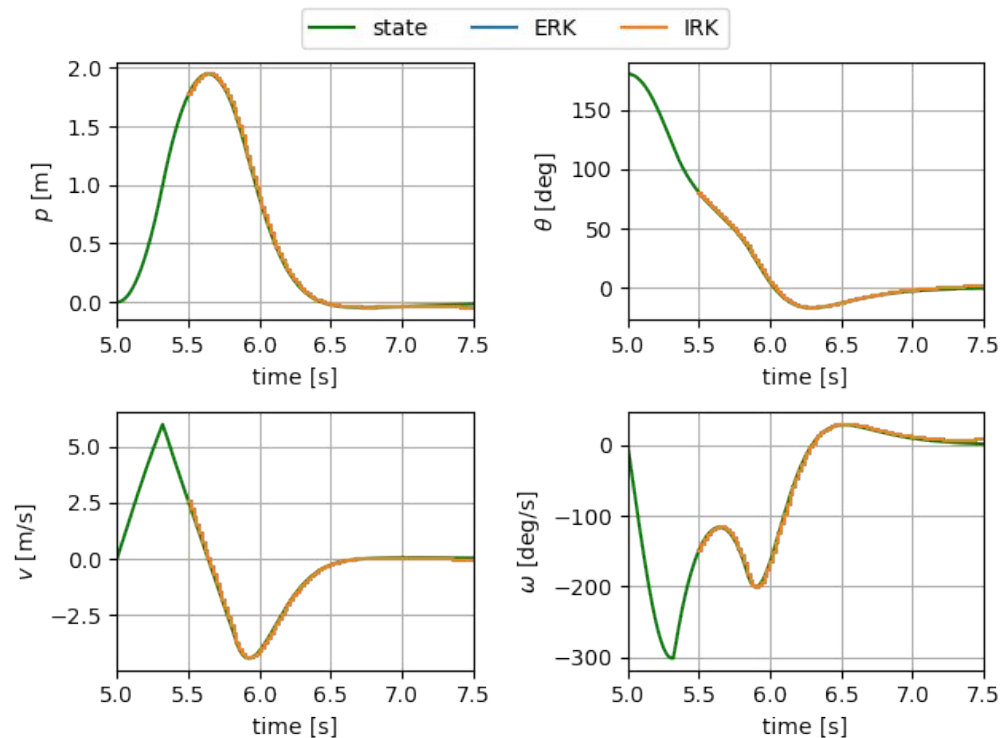
T_s controller = 0.1
 $N = 20$



Controller Integrator Analysis

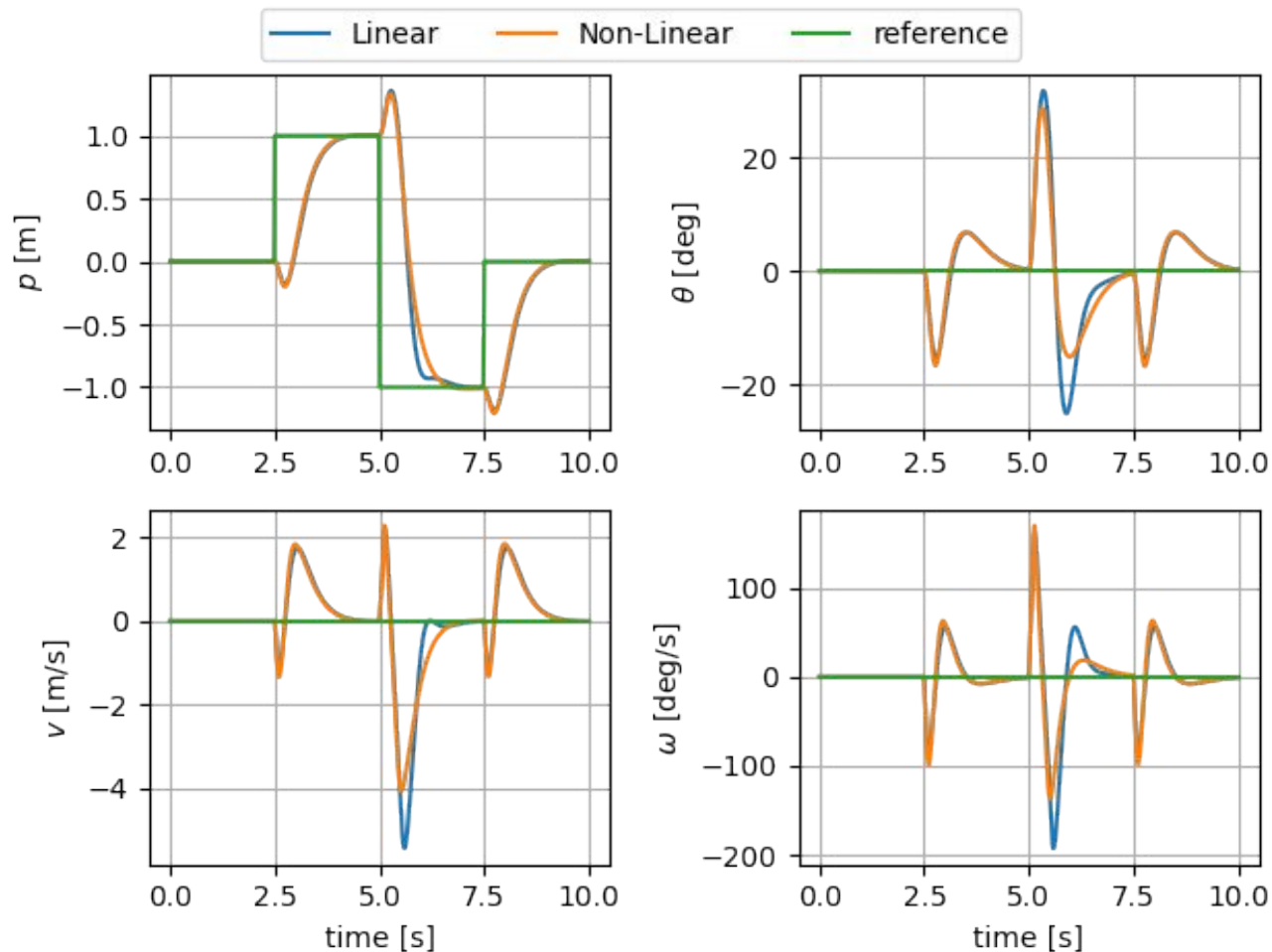
Comparison of the predicted trajectories and mean CPU time
(the proportion is computed per controller integrator)

Mean CPU time	ERK	IRK
$T_s = 0.02$	1	1
$T_s = 0.2$	~ 0.3	~ 0.24



Linear Model

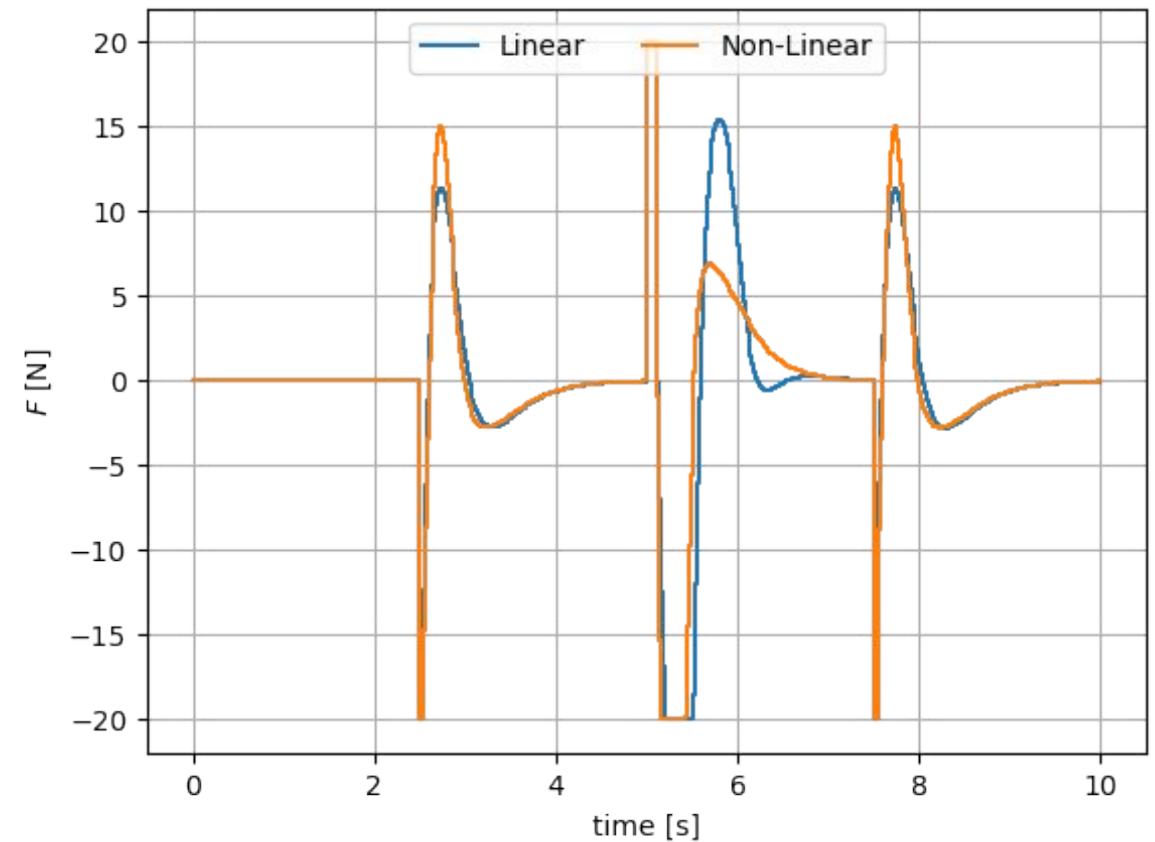
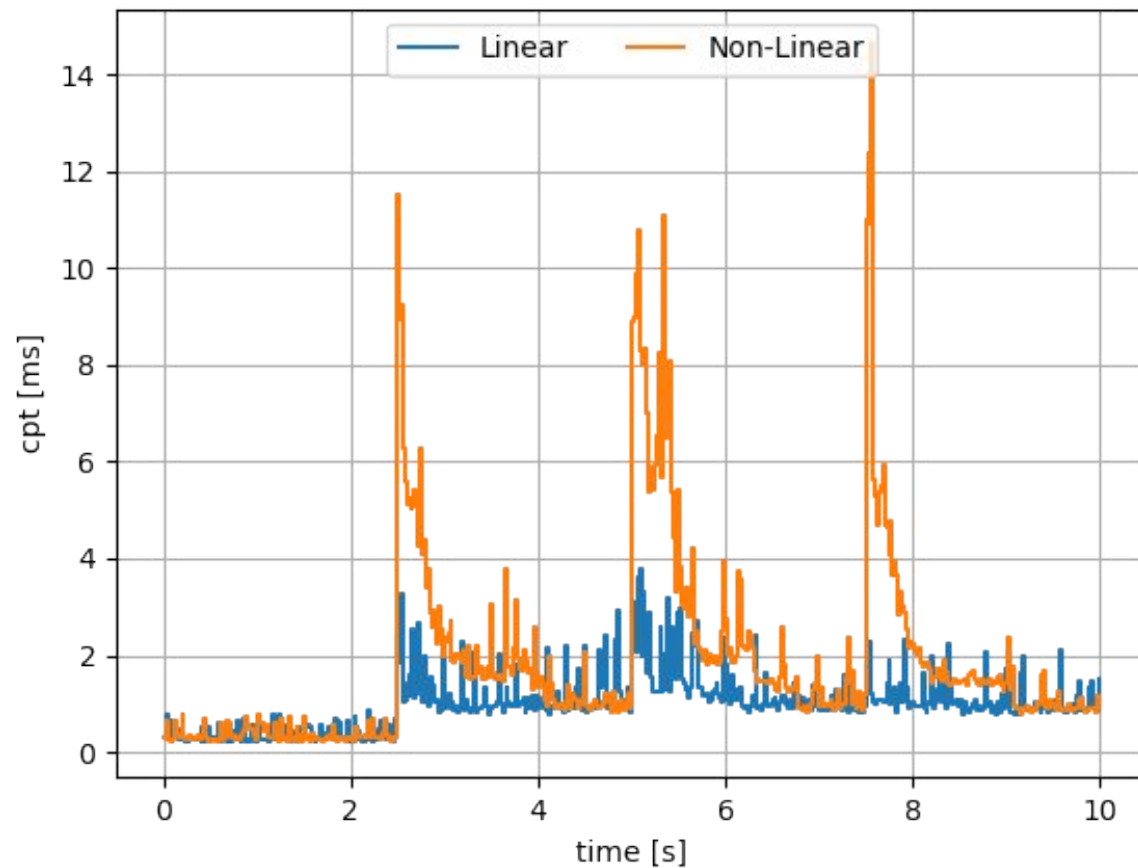
Comparison of the states performances varying model of the controller



	1st step		2nd step	
	Non Linear	Linear	Non Linear	Linear
Overshoot [%]	0.58	0.52	0.67	0.17
Undershoot [m]	0.20	0.18	0.33	0.36
Settling Time [s]	1.55	1.59	1.48	1.61
Underpeak [deg]	16.6	15.3	28.6	31.6
Mean CPU time	1	~ 0.63		

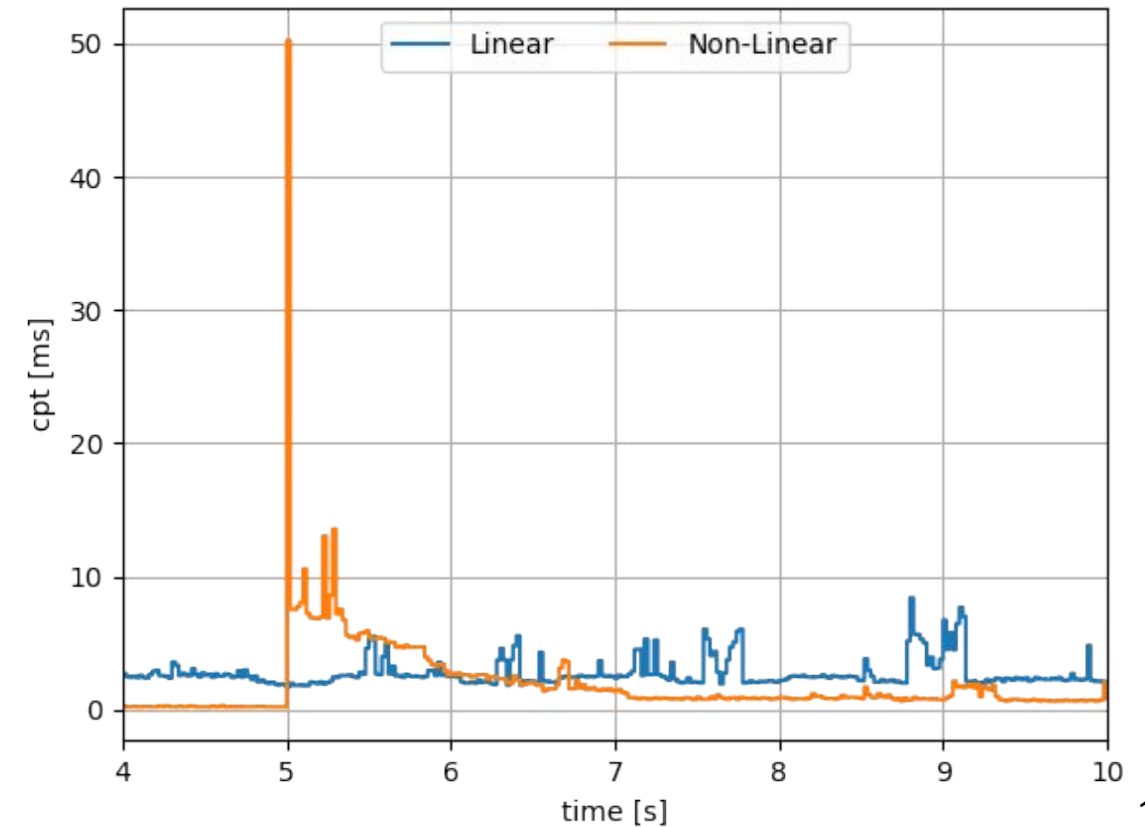
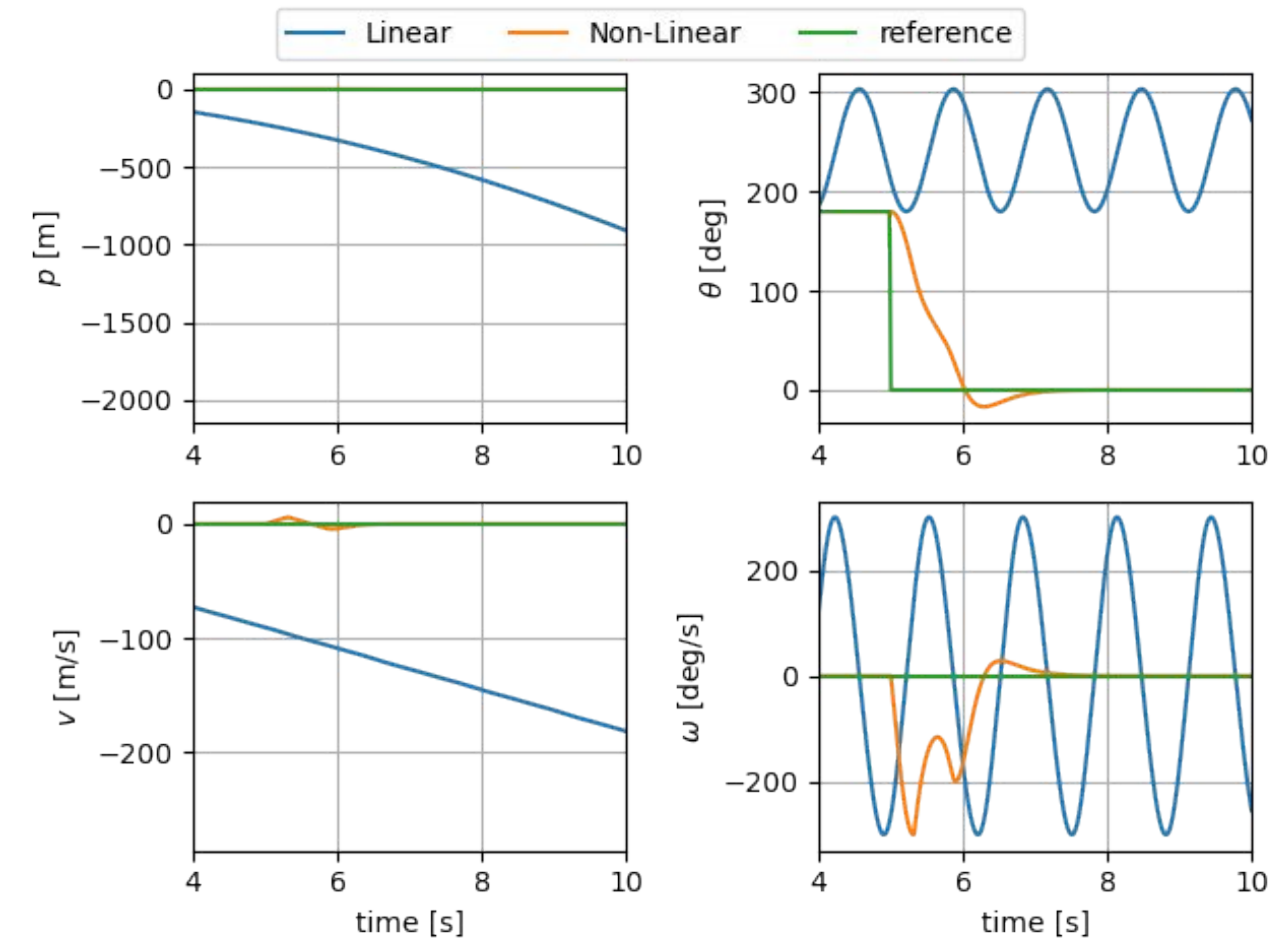
Linear Model

Comparison of the CPU time and the input varying model of the controller

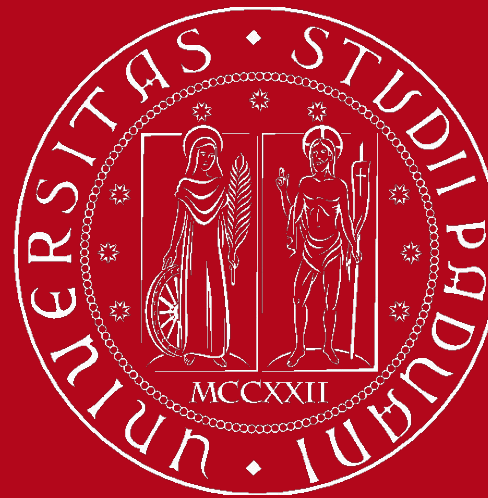


Linear Model

States and inputs of the system trying to track a swing-up maneuver



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